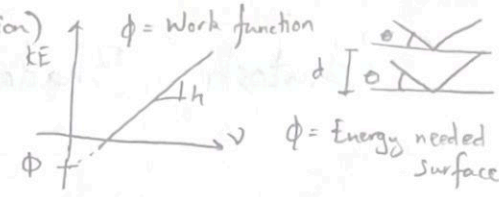
Bohr $E = h\nu = \frac{hc}{\lambda}$; $E = KE + \phi$; $n\lambda = 2d\sin\theta$ (diffraction)

- $\frac{-q^2}{4\pi\varepsilon_0 r_n^2} = -\frac{mv_n^2}{r_n}$; $r_n = \frac{4\pi\varepsilon_0 n^2\hbar^2}{mq^2}$; $V_n = \frac{q^2}{4\pi\varepsilon_0 n\hbar}$
- $KE = \frac{cRh}{n^2}$; $PE = -\frac{2(cR)h}{n^2}$; $(E_{n_2} - E_{n_1}) = cRh\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$



Quantum $\Delta x\,\Delta p \geq \frac{\hbar}{2}$ $\quad \Delta E\,\Delta t \geq \frac{\hbar}{2}$

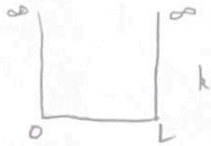
$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2}[V(x) - E]\psi(x)$; $\frac{d\phi(t)}{dt} + \frac{iE}{\hbar}\phi(t) = 0$

$V(x) = 0$: $k_1 = \sqrt{\frac{2mE}{\hbar^2}}$; $\psi_1(x) = Ae^{ik_1x} + Be^{-ik_1x}$

$V(x) \neq 0$: $k_2 = \sqrt{\frac{2m[V(x) - E]}{\hbar^2}}$

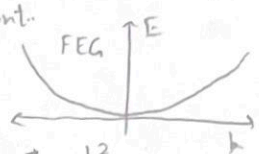
- $E < V_0$: $\psi_2(x) = Ae^{k_2x} + Be^{-k_2x}$
- $V_0 \leq E$: $\psi_3(x) = Ae^{ik_2x} + Be^{-ik_2x}$

BC's: 1) ψ is cont. 2) $\frac{\partial\psi}{\partial x}$ is cont.

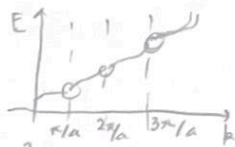


$\psi(x) = C_1\cos k_1x + C_2\sin k_1x$

$k = \sqrt{\frac{2mE}{\hbar^2}}$; $E_n = \frac{\hbar^2k^2n}{2m} = \frac{h^2}{2m}\left(\frac{n\pi}{L}\right)^2$ (k_F = Fermi wave vector)



effective mass $m^* = \frac{\hbar^2}{(\partial^2E/\partial k^2)}$; $\frac{1}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}E^{1/2}$ → Density of states.



Discontinuities due to restricted ψ @ the nucleus

Semiconductors $f(E) \approx \exp\left(-\frac{(E - E_F)}{kT}\right)$; $n_i = N_c\exp\left(-\frac{(E_c - E_F)}{kT}\right)$; $p_i = N_v\exp\left(-\frac{(E_F - E_v)}{kT}\right)$

$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$

If $(E - E_F) > 3k_BT$

- $N_v = 2\left[\frac{2\pi m_pkT}{h^2}\right]^{3/2}$
- $N_c = 2\left[\frac{2\pi m_nkT}{h^2}\right]^{3/2}$
- $\frac{N_v}{N_c} = \frac{m_p^*}{m_n^*}$

$n_0 = n_i\exp\left(\frac{(E_F - E_i)}{kT}\right)$; $p_0 = n_i\exp\left(\frac{(E_i - E_F)}{kT}\right)$; $n_0p_0 = n_i^2$

$n_i^2 = n_0p_0 = \sqrt{N_cN_v}\exp\left(\frac{-E_g}{2kT}\right)$; $\frac{d\mathcal{E}_a}{da} = \frac{q}{\varepsilon}N_d$

$\mathcal{E}(x) = \frac{-dV(x)}{dx}$

$n_i = p_i = n_0 = p_0$
$(E_c - E_F)$
$\simeq (E_F - E_v)$ } Intrinsic

+ Charge carriers $[n, p]$ (ϕ = Flux; J_{net} = current density) $\quad I = J\cdot A$

- $J_{diff} = -qD\frac{\partial n}{\partial x}$; $J_{drift} = \underbrace{q\mu_ic_i\mathcal{E}(x)}_{\phi}$ [for an e^- $q = (-q)$]
- $J_{net} = J_{diff} + J_{drift}$

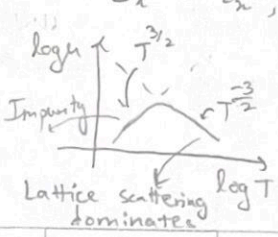
$\langle v_x\rangle = \mu_i\mathcal{E}_x$ (drift velocity) $\quad \{\mu_i = -\mu_n\}$ $\{\mu_i = \mu_p\}$

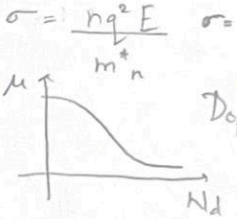
- $J_x = \sigma\cdot\mathcal{E}_x$; $\sigma = \frac{nq^2E}{m_n^*}$ (σ = conductivity) ; $\mu_n = \frac{q\,t}{m_n^*}$ $\quad \mu = \left[\frac{cm^2}{V\cdot s}\right]$ (mean free time)

+ P-N Junction $\frac{D}{\mu} = \frac{kT}{q}$ (Einstein)

$V_b = \frac{k_BT}{q}\ln\left(\frac{P_p}{P_n}\right)$; $N_Ax_{p_0} = N_Dx_{n_0}$

$W = \left[\frac{2V_bkT}{q^2}\ln\left(\frac{N_AN_D}{n_i^2}\right)\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\right]^{1/2}$





Doping:
- $n_0 = n_i + (N_D + N_A)$
- $\rightarrow N_D \gg N_A$ (for n-type)
- $\rightarrow N_D^+ = N_D$ (fully ionized)
- $\rightarrow$ If $N_D \gg n_i$

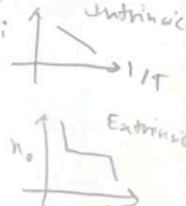
} $n_0 = N_d$

E_F → energy level @ which an e^- has 50% prob. of being occupied at any given time.

$\phi = \frac{\bar{l}^2}{2\bar{t}} = \mu_ic_i\mathcal{E}(x)$

$\frac{d\mathcal{E}_x}{dx} = \frac{\rho}{\varepsilon_s}$ (Poisson equation) (ρ = charge density) $\quad \rho = (N_d - N_A)\cdot q$

pn junction fab:
- Grown junction (abrupt)
- Alloyed junction
- Diffused Junction
- Ion implantation



Constants

$h = 4.14\times10^{-15}$ eV s

$R = 109678$ cm^{-1}

$c = 3\times10^8$ m/s

$\varepsilon_0 = 8.854\times10^{-12}$ C^2/N m^2

$CR = mq^4/8\varepsilon_0^2h^3$

$k_B = 8.68\times10^{-5}$ eV/K $= 1.381\times10^{-23}$ m^2 kg/K s

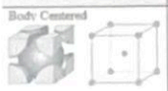
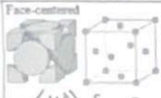
$n_i(Si) = 1.5\times10^{10}$ cm^{-3} [300K]

$m_0 = 9\times10^{-31}$ kg

1 eV $= 1.602\times10^{-19}$ J

Structure	No. of Atoms	Coord. -No.	APF (V_{atoms}/V_{cell})	Side Length
Body Centered	2	8	68%	$a = \frac{4R}{\sqrt{3}}$
Face-centered	4 (eg Ga_1 As_3)	12	74%	$a = 2R\sqrt{2}$
Hexagonal Cubic	6	12 ($c = 2a\frac{\sqrt{2}}{\sqrt{3}}$)	74%	$a = 2R$
Diamond Lattice	8 (eg: Ga_1 As_3 / Si / two FCC's diagonally)	4	34%	$a = \frac{8R}{\sqrt{2}}$

Metals , Ceramics ; Polymers

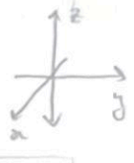
Structure	No. of Atoms	Lattice constants and angles	Coordination number	APF (V_{atoms}/V_{cell})	Side Length	Interstitial (Tetra and then Octa)
Body Centered 	1	a=b=c, α=β=γ=90	8	68%	$a=\frac{4R}{\sqrt{3}}$	(24) (18)
Face-centered  ⟨110⟩ {111} (Close Pack)	4	a=b=c, α=β=γ=90	12	74%	$a=2R\sqrt{2}$	(8) (13)
Hexagonal Cubic  $c=4R\sqrt{\frac{2}{3}}$	6	a=b≠c, α=β=90, γ=120	12	74%	a=2R	(12) (6)

- Principal (n)
- Angular Momentum (ℓ) $0<\ell<n-1$
- Magneti ($2\ell+1$) ↓ (Spatial orientation) No. of energy states for each subshell
- Spin ($\pm 1/2$)

$LD = \frac{\text{No. of atoms}}{\text{length}}$ $PD = \frac{\text{No of atoms}}{\text{Area}}$ Crystal Density ↳ $\rho = \frac{nA}{V_c N_A}$

$n\lambda = 2d\sin\theta$ $d_{hkl} = \frac{1}{\sqrt{h^2+k^2+l^2}}$

* Diff. angle $= 2\theta$





You move your axis according to the req. plane

$E = \frac{-Me^2}{4\pi\varepsilon_0 r}\left(1-\frac{1}{n}\right)$ $E = -\frac{Me^2}{r} + B\frac{e^2}{r^n}$

$U = \frac{-MN_A Z_1 Z_2 e^2}{4\pi\varepsilon_0 r_0}\left(1-\frac{1}{n}\right)$

Crystals $\rho = \frac{n'(A_{cation}+A_{anion})}{V_c N_A}$ ★

Molar lattice energy

$C_1 = \frac{m_1}{(m_1+m_2)}\times 100 = \frac{C_1'A_1}{C_1'A_1+C_2'A_1}\times 100$ $C_1'' = \left(\frac{C_1}{\frac{C_1}{\rho_1}+\frac{C_2}{\rho_2}}\right)$ (Mass/m³)

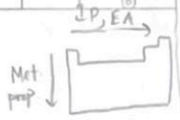
Alloys: $C_1' = \frac{n_{m_1}}{n_{m_2}+n_{m_1}}\times 100 = \frac{C_1A_2}{C_1A_2+C_2A_1}\times 100$

$A_{ave} = \frac{100}{\frac{C_1}{A_1}+\frac{C_2}{A_2}} = \frac{C_1'A_1+C_2'A_2}{100}$

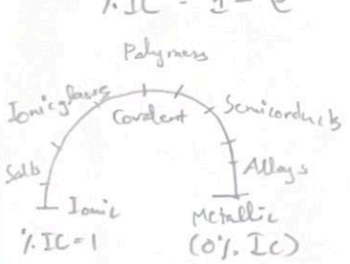
$\rho_{ave} = \frac{100}{\frac{C_1}{\rho_1}+\frac{C_2}{\rho_2}} = \frac{C_1'A_1+C_2'A_2}{\frac{C_1'A_1}{\rho_1}+\frac{C_2'A_2}{\rho_2}}$

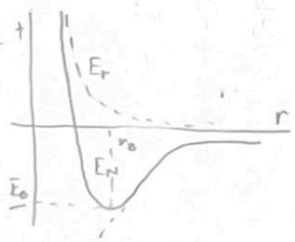
$V_c = a^3$ (BCC ⫸ FCC)

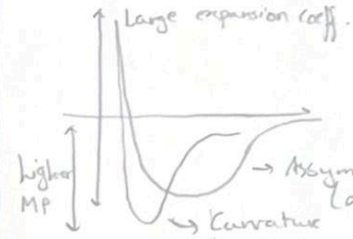
$\chi = \frac{\text{Ionization Pot.} + \text{Electron Aff}}{2}$



$\%IC = 1 - e^{(-0.25(x_a-x_b)^2)}$







Bohr Model
- Discrete orbitals
- Quantized into shells

Wave
- Probability dist.
- 4 quantum no. i.e. quantized into shells & subshells

$E_N = E_A + E_R = \frac{-Z_1Z_2e^2}{4\pi\varepsilon_0 r} + \frac{B}{r^n}$ $n=8$

Z = valence of ion
$e = 1.6\times10^{-19}$ C

{111} = (111); $(\bar{1}\bar{1}\bar{1})$; $(11\bar{1})$; $(\bar{1}11)$; $(1\bar{1}1)$; $(\bar{1}1\bar{1})$; $(1\bar{1}\bar{1})$; $(\bar{1}\bar{1}1)$

Alloys Coord:
- #4 → Zinc Blende (FCC) → Tetrahedral (Anions in)
- #6 → Rock Salt (FCC) → (Octahedral sites) $a = 2(r_{Na}+r_{Cl})$
- #8 → Cesium Cl (BCC) $2r+2R = a\sqrt{3}$

Home-Rotheory:
- ±15% atomic size
- Same crystal structure
- $\Delta E \sim 0$
- Equal or higher in lower

Defects: $N_V = N\exp\left(\frac{-Q_V}{k_B T}\right)$ $N_S = N\exp\left(\frac{-Q_S}{k_B T}\right)$

$Q_S \gg Q_V$ $N = \frac{N_A\rho}{M}$

Burgers: Mag. & Direction of lattice distortion of dislocations

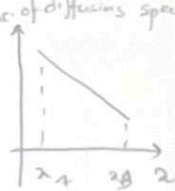
2D: Surface, grain, twins, staking faults

Diffusion (Vacancy vs Interstitial)

① $J_x = -D\frac{dC}{dx}$ (flux (kg/m².s); Diff. Coeff.)

$\frac{dC}{dx} = \frac{\Delta C}{\Delta x} = \frac{C_2-C_1}{x_2-x_1}$

$D = D_0\exp\left(\frac{-Q_d}{RT}\right)$



- Diffusing species (Atomic size) (Self-diffusion vs interstitial)
- Temperature

② $\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$

Frenkel defect: Cation Vacancy + Cation interstitial

Schottky Defect: Cation Vacancy + Anion Vacancy

MSE Midterm 1 Review

Chapter 1

+ Virial theorem $\left(PE = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r}\right)$; $\overline{KE} = -\frac{\overline{PE}}{2}$; $\overline{E}_{tot} = \frac{1}{2}\overline{PE}$

Ex. 1.1 Pg 7

$\text{Force} = \frac{dE}{dr}$

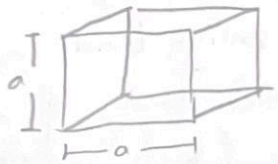
$\overline{KE} = \frac{1}{2}mv^2$; $v = \frac{2\pi r_0}{T}$

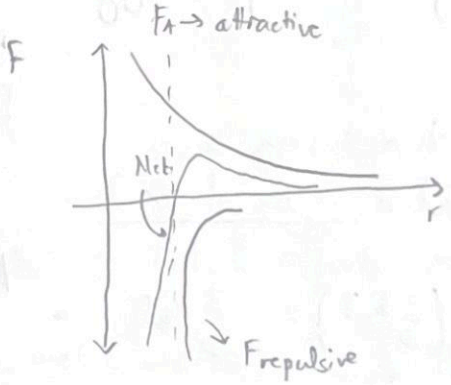
orbital frequency $\rightarrow \nu = \frac{1}{T}$

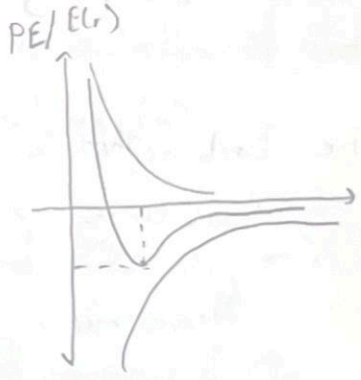
+ $\left.\frac{dE}{dr}\right|_{r_0} = 0$

+ $PE_{binding} = PE_{ionic} - PE_{charges}$

+ $\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{\#\text{ Atoms}}{\text{unit cell}} \cdot \frac{1\text{ mol}}{6.023\times10^{23}\text{ atoms}} \times \frac{M^w\ g/mol}{a^3} \times \frac{kg}{1000\ g}$







+ Gases and Heat Capacity (1.4)

@ constant volume, $\rightarrow \boxed{C_m} = \frac{dU}{dT} = \frac{3}{2}N_A k = \frac{3}{2}R$ [1.13]

Where $\boxed{U}$ $N_A\left(\frac{1}{2}m\overline{v^2}\right) = \frac{3}{2}N_A kT$

$\boxed{U}$ → Total internal energy of monoatomic gas

The rise in the internal energy per unit temperature is called the heat capacity. $[C_m]$

Kinetic Molecular Theory: Applying classical equations of motion to gases.

[1.10] Theoretically: $PV = \frac{1}{3}Nm\overline{v^2}$; Experimentally: $PV = \left(\frac{N}{N_A}\right)RT$

Comparing these two

We obtain,

$$PV = \frac{2}{3}N\left(\frac{1}{2}m\overline{v^2}\right) \qquad \frac{1}{3}m\overline{v^2} = \frac{R}{N_A}T$$

(same as theoretical)

where $\overline{KE} = \frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$ ★

(mean square velocity $\propto$ absolute temperature)

Assumptions,

1) Molecules in constant random motion
2) Intermolecular forces are negligible
3) Volume of gas molecules is negligible
4) Elastic collisions
5) Newtonian mechanics can be applied

$$\boxed{U = N_A\left(\frac{1}{2}m\overline{v^2}\right) = \frac{3}{2}N_A kT} \quad ; \quad \boxed{C_m = \frac{dU}{dT} = \frac{3}{2}R}$$

Maxwell's principle of equipartition of energy → assigns an average of $\frac{1}{2}kT$ to each independent energy term in the expression for the total energy of a system

i.e. Each method in which a system can absorb energy is called a <u>degree of freedom</u>

Monoatomic: $3\left(\frac{1}{2}k_BT\right) = \frac{3}{2}k_BT$ → Average $\overline{KE}$

Diatomic (O_2): 3 translational + 2 rotational = $5\left(\frac{1}{2}k_BT\right)$

$k_B = \frac{R}{N_A}$

For <u>solids</u>: atoms bonded as "springs"

$U = 3RT$; $\boxed{C_m = \frac{dU}{dT} = 3R}$ → Dulong-Petit Rule

[=] J K^{-1} mol^{-1}

Failures of KE Theory (Cannot predict):
+ Resistivity of a metal increases linearly w/ absolute temp.
+ At low T, heat capacity increases as T^3

2. Drude Model (Electrical & Thermal Conduction)

[2.2] $J = \frac{\Delta q}{A \Delta t} = \frac{I}{A}$ for e^- $\longrightarrow$ $J_x = e n v_{d_x}$

(As e^- carry current)

↓

Conduction electrons. (They have both PE and KE) and move randomly in lattice w/ no net displacement

↓

Drift velocity

τ → mean scattering time (mean time between collisions) $l = v_F \cdot \tau$

→ average velocity of e^- in the x-direction

[2.4] $\boxed{v_{d_x} = \mu_d E_x}$; $\boxed{\mu_d = \frac{e\tau}{m_e}}$

↑ Drift mobility

Conductivity

↓

This therefore gives $J_x = e n \mu_d E_x$ and $\sigma = e n \mu_d$ [2.6, 2.7]

→ How fast e^- will drift when driven by an applied field

Low e^- scattering → $\tau \uparrow$, $\mu_d \uparrow$ but σ maybe ↓ or ↑ because it also depends on the concentration of cond. electrons (n)

+ Temperature dependence

Thermal vibration scattering → affects τ (which affects μ_d and eventually σ)

$\rho_T = AT$ } Pure metal resistivity due to thermal vibrations of the crystal

↑

lattice-scattering-limited conductivity.

Electron scattering can also be affected by lattice (& distortions due to impurit

$$\frac{1}{\tau} = \frac{1}{\tau_T} + \frac{1}{\tau_I} \quad ; \quad \frac{1}{\mu_d} = \frac{1}{\mu_L} + \frac{1}{\mu_I} \quad ; \quad \rho = \rho_T + \rho_I = \frac{1}{en\mu_L} + \frac{1}{en\mu_I}$$

ρ_I has little temp. dependence (can be clubbed under ρ_R)
and therefore $\rho(T) = AT + B$

Matthiessen's rule (summation of resistivities)

$$\alpha_0 = \frac{1}{\rho_0}\left[\frac{d\rho}{dT}\right]_{T=T_0}$$

α_0 is assumed constant over a temp. range T_0 to T to give: $\boxed{\rho = \rho_0[1 + \alpha_0(T - T_0)]}$

ρ_0, α_0 depend on reference temp. T_0

Exceptions → Magnetic materials and alloys (dominated by residual resistivity due to lattice)

↓

Less temp. dependence as $\rho_R(\rho_I) > \rho_T$

Nordheim's Rule (single phase)

Important Semi-empirical equation to predict the resistivity of an alloy

Solid solution $\rho_I = Cx(1-x)$

Assumes: 1) Solid solution has the solute atoms randomly distributed in the lattice

2) For dilute solut, $\rho_t = Cx$

3) Solute atom that is drastically different in size to the solvent atoms result in larger ρ_I & larger C.

4) Alloying does not significantly vary the no. of conduction e^- per atom in the alloy

$\therefore \rho = \rho_{matrix} + \rho_I$

$= (\rho_T + \rho_R) + Cx(1-x)$

- ρ_T: Temp-dependence
- ρ_R: defects, grain boundaries
- $Cx(1-x)$: Impurities

(2.4)

Effective resistivity (multi-phase solutions)

Resistivity Mixture rule (for similar species)

$$\boxed{\rho_{eff} = x_\alpha \rho_\alpha + x_\beta \rho_\beta}$$

$x_\alpha = \frac{L_\alpha}{L}$ (L_α: length of α layers; L: length of sample)

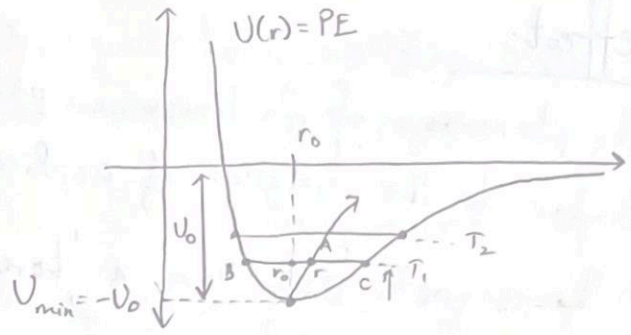
For different species:

If $\rho_{dispersed} > 10\rho_{continuous}$, $\rho_{eff} = \rho_c\frac{(1 + \frac{1}{2}x_d)}{(1 - x_d)}$; $\rho_d < 0.1\rho_c \Rightarrow \rho_{eff} = \rho_c\frac{(1 - x_d)}{(1 + 2x_d)}$

Thermal Expansion

Why do materials expand on heating?

This is due to the asymmetric nature of the PE curve.



+ All particles vibrate along their mean position (w/ total energy given by $E = U + KE$) (w/ overall E as constant when the atom is vibrating in SHM)

+ B-A-C are states of the vibrating atoms, A being when KE ↑ and B & C when U ↑ ($K_E = 0$). → BC corresponds to the total energy at that temperature.

+ However as the U(r) curve is asymmetric, the atoms spend more time in $r > r_0$ therefore increasing the average separation

+ As T ↑↑, E ↑ and the atoms vibrate between wider extrema → increasing average separation

$$L = L_0[1 + \lambda(T - T_0)] \qquad [1.15]$$

↳ thermal expansion coefficient $\lambda = \lambda(T)$

Hall effect:

+ The phenomena of a transverse field when a magnetic field is applied perpendicular to the direction of applied field.

+ The magnetic field causes a 'Lorentz force' on each electron given by

$$F = qv \times B \quad [2.29]$$

+ As the charges accumulate, there is an internal electric field → Hall f

+ As the charges accumulate, there is an internal electric field → Hall voltage, which gives rise to Hall voltage. The charge keeps accumulating until the e^- start to get repelled (ie. E_H is big enough so the e^- can't climb

@ SS:

$$eE_H = ev_d B_z \quad ; \quad J = env_d \quad ; \quad v_e = \frac{\mu_e}{e} F_{net}$$

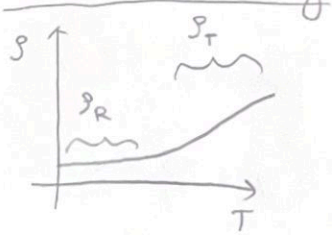
$n = \frac{\sigma}{e\mu_d}$ ↑ no. of conduction e^-

$$E_H = \frac{J}{en} B_z \quad [2.30]$$

negative sign because

$$\frac{1}{en} = \frac{E}{(J B_z)} \quad ; \quad \boxed{R_H = -\frac{1}{en}}$$

Metals & Resistivity

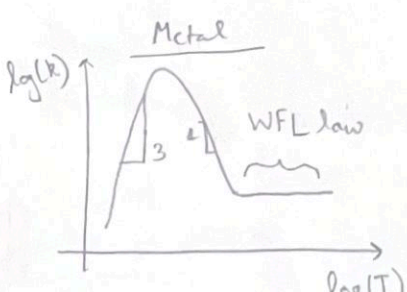


$\rho = \rho_T + \rho_I + \rho_R$ (Lattice Imp. / dislocations)

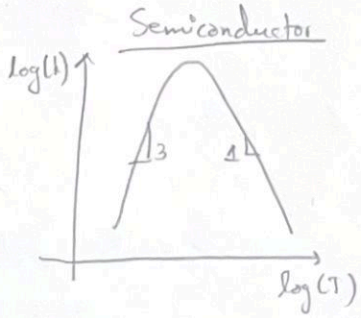
Scattering limited conductivity

$\rho_T = AT + B$; ρ_I and ρ_R do not have temp. dependence. Therefore these values will only shift the graph up.

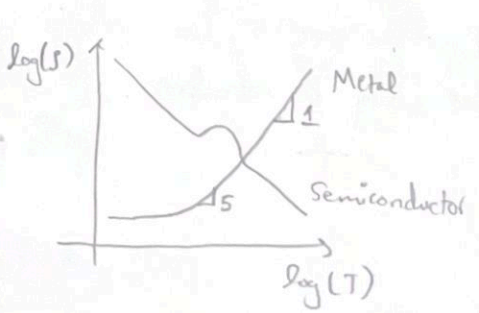
$\rho_I = C x(1-x)$ (Nordheim's Rule)



$$\frac{1}{\sigma T} = C_{WFL}$$



Phonon mediated



Chapter 3

Electromagnetic wave → Electric & Magnetic fields are perpendicular to each other & direction of propagation x

[3.1] $E_y(x,t) = E_0 \sin(kx - \omega t)$; $c = \frac{\omega}{k} = \nu\lambda$

[3.2] $I = \frac{1}{2} c\varepsilon_0 E_0^2$ → Classical light intensity

[3.3] $2d\sin\theta = n\lambda$ → Bragg's law

The wave nature of light is exhibited by Young's double slit experiment

(i) Constructive interference

(ii) Destructive interference

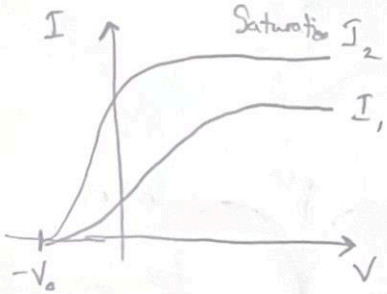
↳ Photoelectric effect :

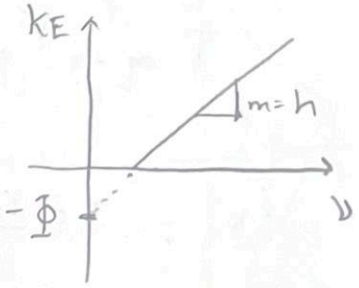
Uses the particle-like nature of particles to describe the emission of e^- from surface of a cathode.

$$KE = h\nu - h\nu_0 \qquad E = h\nu_0 = \frac{hc}{\lambda_0} \qquad [3.4]$$

The saturation current ∝ light intensity (i.e more e^- are emitted)

(However, KE_m of emitted e^- is independent of the light intensity I





Photon flux → $\Gamma_{ph} = \frac{\Delta N_{ph}}{A\,\Delta t}$ [3.5]

$I = \Gamma_{ph} h\nu$ [3.6]

I ← Light intensity

↳ Compton Scattering :

Electron is scattered when incident by a photon (EM waves)

Therefore the photon must have a momentum given by:

$J = e\Gamma_{ph}$

$p = \frac{h}{\lambda}$; $E = h\nu = \hbar\omega$ } Photon energy & momentum

$p = m_e v_{electron}$ → momentum of an e^-

De Broglie Relationship

Electrons behave in a similar way as photons.

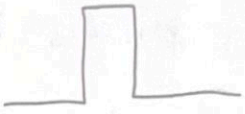
[3.13] $\lambda = \frac{h}{p}$ $\quad KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$; $\boxed{p = \sqrt{2m\,KE}}$

Heisenberg: $\Delta x \Delta p_x \geq \hbar$; $\Delta E \Delta t \geq \hbar$

$(\Delta x \Delta k \geq 2\pi)$ $\quad (\Delta t \Delta \omega \geq 2\pi)$

Quantum

TISE: [3.16a] $\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E-V)\psi = 0$



BC:
1) ψ is continuous
2) $\frac{d\psi}{dx}$ is continuous

Solutions: $\psi''(x) + k^2\psi(x) = 0$ $\quad \psi''(x) - \kappa^2\psi(x) = 0$ $\quad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$

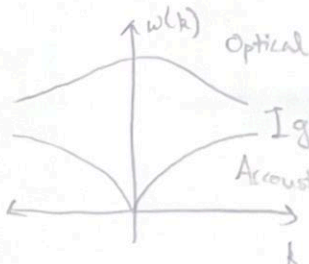
$\psi_{II}(x) = \psi_I(x) = Ae^{-ikx} + e^{+ikx}$ $\quad \psi_{II}(x) = Ce^{\kappa x} + De^{-\kappa x}$

$k = \frac{2mE}{\sqrt{\hbar^2}}$

T eq → Pg 223

More on phonons $\quad E = \hbar\omega$; $p = \hbar k$; $k = \frac{2\pi}{\lambda}$

+ Transverse & longitudinal
+ When we have diatomic → we start to get interactions
  - ↳ Acoustic: neighbouring atoms in phase → long wavelength
  - ↳ Optical: " out of phase → shorter wavelength



→ optical modes have more lattice distortion compared to acoustic phonon Δ need more energy to excite (higher freq)

→ Long. have higher Degree of freedom

Dispersion relations

